

Nexora Commerce

Retail Sales, Customer & Profitability Analytics

Tools: Microsoft Excel, Power Query, Microsoft Power BI, DAX

Analysis Period: January 2024 – December 2025

Project Type: Data Analytics Portfolio Project

Dataset Type: Synthetic Retail Dataset

1. Executive Snapshot

Nexora Commerce is a fictional omnichannel retail business operating across multiple regions in India through online and physical store channels.

The objective of this project is to analyze sales, profitability, customer performance, product performance, discount levels, and regional performance to identify the factors affecting business profitability.

The analysis shows that Nexora generates substantial sales but remains overall loss-making.

Key Performance Indicators

- **Total Sales:** ₹203.15M
- **Total Cost:** ₹217.08M
- **Total Profit:** -₹13.94M
- **Profit Margin:** -6.86%
- **Total Orders:** 20K
- **Total Customers:** 4K
- **Total Quantity Sold:** 10K
- **Average Order Value:** ₹10K

The findings indicate that revenue generation alone is not translating into profitability. High-sales products, higher discount levels, regions, customer segments, and sales channels all require profitability-focused monitoring.

2. Company & Business Context

Nexora Commerce is a fictional retail and e-commerce company serving individual consumers through both online and physical store channels.

The company sells products across multiple categories, including:

- Electronics
- Home & Kitchen
- Fashion
- Beauty & Personal Care
- Sports & Fitness

The business operates across four major regions:

- South
- West
- North
- East

The management team wants to understand why strong sales are not translating into positive profitability.

3. Business Problem

Although Nexora Commerce generates significant sales revenue, the business is experiencing negative overall profitability.

Management wants to understand:

- Which products and categories are generating losses?
- Are higher discounts associated with weaker profit margins?
- Which regions contribute most to sales and losses?
- Which customer segments generate the most revenue?
- Are high-sales customers actually profitable?
- How do online and store channels compare?
- Which areas should management focus on to improve profitability?

The primary business concern is therefore revenue quality rather than sales volume alone.

4. Business Objective

The main objective of this project is to analyze retail sales and profitability data and identify the major factors associated with negative profitability.

The analysis focuses on:

1. Sales performance
 2. Product profitability
 3. Customer profitability
 4. Regional performance
 5. Discount impact
 6. Sales channel performance
 7. High-sales loss-making products
 8. Business recommendations for improving profitability
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5. Business Questions

Sales Performance

- How are sales changing over time?
- Which categories and subcategories generate the most sales?

- Which regions contribute the most sales?
- Which sales channel generates more revenue?

Profitability

- Which categories and products are loss-making?
- Which products generate high sales but negative profit?
- Which regions have negative profitability?
- What is the overall profit margin?

Customer Performance

- Which customer segments generate the most sales?
- Are high-sales customers profitable?
- How does customer profitability vary across segments?

Discount Analysis

- How does profitability vary across discount levels?
- Are higher discounts associated with weaker profit margins?
- Do higher discounts consistently result in higher sales?

Channel Performance

- How do Online and Store channels compare?
- Does the channel generating higher sales also generate better profitability?

6. Dataset & Data Preparation

The project uses a synthetic retail dataset designed to simulate a realistic business environment.

The dataset contains four related tables:

Customers

Contains customer-level information such as:

- Customer ID
- Customer Name
- Gender
- Age
- City
- State
- Region
- Customer Segment
- Registration Date

Products

Contains product information such as:

- Product ID
- Product Name
- Category
- Subcategory
- Supplier
- Unit Cost
- Selling Price

Orders

Contains order-level information such as:

- Order ID
- Order Date
- Customer ID
- Sales Channel
- Sales Platform
- City

- Region
- Payment Method
- Order Status

Order Details

Contains transaction-level product information such as:

- Order ID
 - Product ID
 - Quantity
 - Unit Price
 - Discount
 - Sales Amount
 - Cost Amount
 - Profit
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7. Data Quality Assessment

Several data quality issues were intentionally introduced into the synthetic dataset to simulate a realistic analytics workflow.

The identified issues included:

- Missing customer segments
- Missing city values
- Missing supplier information
- Inconsistent category capitalization
- Zero-quantity transaction records
- Duplicate transaction-detail rows
- High-discount records
- Negative-profit transactions
- Text formatting inconsistencies

These issues were reviewed before analysis.

8. Data Cleaning & Transformation

Data cleaning was performed using Power Query in Excel/Power BI.

Customers

- Missing Customer Segment values were replaced with `Unknown`.
- Missing City values were replaced with `Unknown`.
- Text fields were trimmed.
- Data types were checked.
- Redundant fields were removed where appropriate.

Products

- Missing Supplier values were replaced with `Unknown`.
- Category names were standardized.
- Text formatting was cleaned.
- Data types were checked.

Orders

- Duplicate Order IDs and blank records were checked.
- Text fields were trimmed.
- Data types were validated.
- No major duplicate or blank Order ID issues were identified.

Order Details

- Eight zero-quantity records were removed.
- Exact duplicate rows were removed.

- Negative-profit records were retained because they represent meaningful business outcomes.
 - Discounts, including 50% discount records, were retained for analysis.
 - Financial fields were rounded to two decimal places.
 - Text and ID fields were trimmed.
 - Data types were validated.
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9. Data Model & Analytical Approach

The cleaned tables were loaded into Power BI and connected using relationships between key identifiers.

Relationships

- Customers → Orders
- Orders → Order Details
- Products → Order Details

The analytical model allows sales, costs, profits, customers, products, regions, channels, and discounts to be analyzed together.

DAX measures were created for the major business KPIs, including:

- Total Sales
 - Total Cost
 - Total Profit
 - Profit Margin %
 - Total Order
 - Total Customers
 - Total Quantity
 - Average Order Value
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10. Key Findings

10.1 Overall Profitability

Nexora Commerce generated approximately **₹203.15M in sales**, while total cost was approximately **₹217.08M**.

This resulted in:

- **Total Profit:** -₹13.94M
- **Profit Margin:** -6.86%

The business therefore generates substantial revenue but remains loss-making overall.

10.2 Category Performance

All five major product categories are loss-making.

Approximate category-level profit:

Category	Total Profit
Fashion	-₹13.89M
Electronics	-₹26.83M
Beauty & Personal Care	-₹31.29M
Home & Kitchen	-₹33.06M
Sports & Fitness	-₹34.29M

Fashion has the highest Total Profit among the categories, but this still represents a loss of approximately ₹13.89M.

At the subcategory level, Footwear has the smallest loss at approximately **₹2.57M**.

11. High-Sales Loss-Making Products

Several products generate substantial sales while still producing negative profit.

Three high-sales loss-making products identified were:

Accessories Product 002

- Sales: ₹6,582,697.58
- Cost: ₹7,182,552.19
- Profit: -₹599,854.60
- Profit Margin: -9%
- Average Discount: 19%

Accessories Product 006

- Sales: ₹6,350,110.32
- Cost: ₹6,892,253.55
- Profit: -₹542,143.23
- Profit Margin: -9%
- Average Discount: 19%

Personal Care Product 106

- Sales: ₹6,271,032.22
- Cost: ₹6,615,013.50
- Profit: -₹343,981.26
- Profit Margin: -5%
- Average Discount: 19%

Together, these three products generate approximately **₹1.92 crore in sales** while producing approximately **₹14.86 lakh in losses**.

This demonstrates why sales volume alone should not be used to evaluate product performance.

12. Discount & Profitability Analysis

Profitability becomes increasingly negative as discount levels increase.

Discount	Profit Margin
15%	-2.2%
20%	-5.3%
25%	-7.9%
30%	-13.6%
50%	-54.8%

The analysis shows an association between higher discount levels and weaker profit margins.

However, this analysis does **not establish that discounts alone caused the losses**.

The relationship should therefore be interpreted as an observed pattern rather than a causal conclusion.

Sales also do not consistently increase as discount levels become higher.

13. Regional Performance

All four regions are loss-making.

Region	Sales	Profit	Profit Margin
East	₹30.38M	-₹1.93M	-6.34%
North	₹44.04M	-₹3.02M	-6.85%
West	₹59.90M	-₹4.16M	-6.94%
South	₹68.81M	-₹4.83M	-7.03%

South generates the highest sales and also has the largest absolute loss.

This indicates that high regional sales do not necessarily translate into stronger profitability.

14. Customer Segment Performance

Customer segment analysis shows negative profitability across the identified segments.

Approximate profit margins:

- Consumer: -6.6%
- Corporate: -6.9%
- Small Business: -7.0%
- Unknown: -8.5%

Consumer customers generate the largest share of sales, approximately **₹140M**, but profitability remains negative.

This indicates that customer revenue contribution should be evaluated together with profitability.

15. Sales Channel Performance

The business operates through Online and Store channels.

Online

- Sales: approximately ₹133M
- Profit Margin: approximately -6.8%

Store

- Sales: approximately ₹71M
- Profit Margin: approximately -6.9%

Online generates substantially higher sales, but both channels have similarly negative margins.

Therefore, higher sales through the Online channel have not translated into meaningfully better profitability.

16. Drivers Behind the Profitability Problems

Driver 1 — Pricing and Discount Pressure

Loss-making products show substantial discounts.

For example, Accessories Product 002 has:

- Average Cost: ₹38,409
- Average Price: ₹43,293
- Average Discount: 19%

Although the average price is above the average cost before discount, applying discounts reduces the realized selling price and creates additional margin pressure.

Similar patterns are visible for other high-sales loss-making products.

Driver 2 — Revenue Growth Without Sufficient Profitability

Driver 3 — Broad-Based Profitability Weakness

Negative profitability is observed across:

- Major product categories
- Regions
- Customer segments
- Online channel
- Store channel

Therefore, the available analysis does not support a single explanation for the overall losses.

Further investigation of product costs, supplier pricing, transaction-level discounts, pricing policies, and product mix would be required.

17. Business Impact

17.1 Revenue Growth Is Not Translating Into Profit

The business generates approximately ₹203.15M in sales but records a loss of approximately ₹13.94M.

This indicates that revenue generation is not currently producing sufficient financial returns.

17.2 High-Sales Products Can Increase Losses

Products generating significant sales can still negatively affect overall profitability when their costs and discounts are not managed effectively.

17.3 Higher Discounts Create Margin Pressure

Higher discount levels are associated with increasingly negative profit margins.

If discounts are not evaluated against product cost and target margins, they can create significant margin pressure.

17.4 Regional Performance Requires Profitability Monitoring

All regions are loss-making.

South has the highest sales as well as the largest absolute loss, showing that regional sales volume alone is not enough to evaluate performance.

17.5 Customer and Channel Revenue Does Not Guarantee Profitability

Consumer customers and Online sales contribute substantial revenue, but both remain loss-making.

The business therefore needs to monitor customer and channel profitability alongside revenue.

18. Management Recommendations

Recommendation 1 — Review High-Sales, Loss-Making Products

Management should investigate high-sales loss-making products such as:

- Accessories Product 002
- Accessories Product 006
- Personal Care Product 106

Possible actions include:

- Reviewing selling prices
- Negotiating supplier costs
- Reducing unnecessary discounts
- Reviewing product mix
- Evaluating whether certain products should continue to be promoted

Recommendation 2 — Introduce Margin-Based Discount Controls

Discount decisions should consider product cost and target profitability.

Management could:

- Establish category-specific discount limits
- Monitor high-discount transactions
- Review products where discounts significantly reduce margins
- Evaluate discounts based on incremental profitability rather than sales volume alone

Recommendation 3 — Review Pricing and Product Cost Structure

The company should evaluate:

- Unit cost
- Selling price
- Realized selling price after discounts
- Profit margin
- Supplier pricing
- Product mix

Supplier negotiations and pricing adjustments could be considered where appropriate.

Recommendation 4 — Monitor Regional Performance Using Profitability KPIs

Regional performance should be monitored using:

- Sales
- Total Profit
- Profit Margin
- Discount levels

- Product mix

South requires particular investigation because it generates the highest sales and largest absolute loss.

Recommendation 5 — Evaluate Customers and Channels Based on Profitability

Customer segments and sales channels should be evaluated using both revenue and profitability metrics.

Management should monitor:

- Sales
- Profit
- Profit Margin
- Order value
- Discount levels

This would help identify revenue that contributes to sustainable business performance.

19. Recommended Management Focus

The analysis suggests that management should focus on improving revenue quality rather than simply increasing sales volume.

A practical approach would be:

Identify loss-making areas → Investigate pricing, cost and discount drivers → Apply targeted corrective actions → Monitor profit and margin alongside sales

The dashboard can then be used as an ongoing monitoring tool for these profitability KPIs.

20. Dashboard Structure

The Power BI dashboard consists of three analytical pages.

Page 1 — Executive Overview

Provides a management-level summary of:

- Sales
- Orders
- Customers
- Quantity
- Cost
- Profit
- Average Order Value
- Profit Margin
- Sales trend
- Profit trend
- Regional performance
- Channel performance
- Key business takeaways

Page 2 — Product & Profitability

Focuses on:

- Category and subcategory performance
- Discount versus profit margin
- High-sales loss-making products
- Subcategory profitability
- Top loss-making products

Page 3 — Customer Analysis

Focuses on:

- Customer segment performance
 - Top customers by sales
 - Customer sales versus profitability
 - Customer profitability insights
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21. Limitations

This project has several limitations:

- The dataset is synthetic and does not represent actual company data.
 - The analysis is limited to the variables available in the dataset.
 - Competitor pricing and market conditions are not included.
 - Customer acquisition costs are not included.
 - Supplier market conditions are not available.
 - The relationship between discount levels and profitability does not establish causation.
 - Further investigation is required to identify the exact operational and financial causes of the losses.
 - The analysis covers the period from January 2024 to December 2025.
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22. Challenges & Learning

During the project, several practical data analytics challenges were addressed.

Challenges

- Handling missing values across related datasets
- Identifying duplicate and invalid transaction records

- Removing zero-quantity transactions
- Retaining meaningful negative-profit transactions
- Standardizing inconsistent category values
- Validating data types
- Building relationships between multiple tables
- Creating DAX measures
- Designing visuals around business questions
- Avoiding unsupported causal conclusions
- Translating analytical findings into business recommendations

Key Learning

The project provided practical experience in:

- Excel data handling
- Power Query data cleaning
- Data modelling
- Power BI dashboard development
- DAX calculations
- KPI analysis
- Profitability analysis
- Customer analysis
- Product analysis
- Business-focused data storytelling
- Converting analytical findings into actionable recommendations

23. Project Outcome

The Nexora Commerce project demonstrates an end-to-end data analytics workflow:

Business Problem → Data Understanding → Data Quality Assessment → Data Cleaning →
Data Modelling → DAX Analysis → Dashboard Development → Findings → Business
Impact → Recommendations

The project demonstrates how a data analyst can move beyond reporting sales numbers and use profitability, customer, product, regional, channel, and discount analysis to identify business problems and support management decision-making.

Tools Used:

Microsoft Excel | Power Query | Microsoft Power BI | DAX

Project:

Nexora Commerce — Retail Sales, Customer & Profitability Analytics